

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I **Kasif jalal M (192424214)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Kasif jalal M

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.
4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

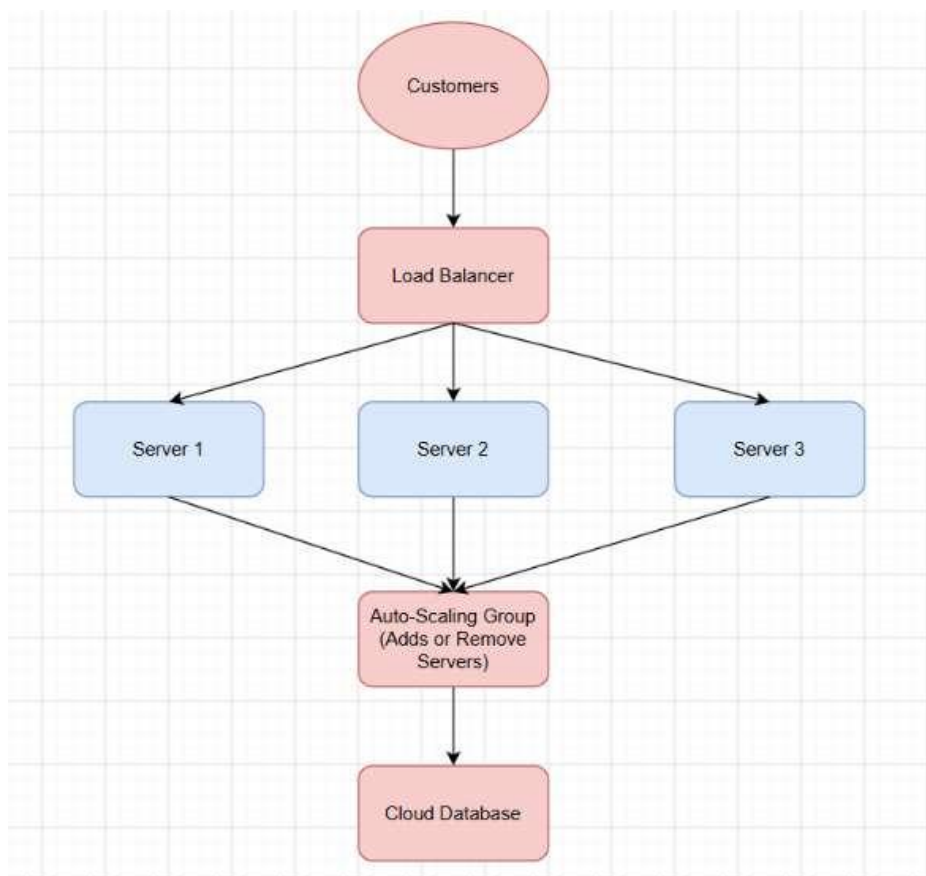
1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario

Handling Unpredictable Traffic with Auto-Scaling and Load Balancing

During seasonal sales, an e-commerce company may experience a sudden increase in website visitors. If the infrastructure cannot handle the extra traffic, customers may face slow loading pages, failed payments, or website crashes. Cloud technologies such as auto-scaling and load balancing help solve these problems.

- **Load Balancing** distributes incoming customer requests evenly across multiple application servers.
- This prevents any single server from becoming overloaded.
- **Auto-Scaling** automatically increases the number of servers when traffic increases and removes extra servers when demand decreases.
- Together, these technologies ensure that the application remains responsive even during peak shopping periods.

Architecture Diagram:



How This Approach Reduces Downtime and Improves User Experience

Reducing Downtime

- If one server becomes overloaded or fails, the load balancer redirects requests to healthy servers.
- Auto-scaling launches additional servers automatically when website traffic increases.
- During low traffic, unnecessary servers are removed to reduce costs without affecting performance.
- Since the system adjusts automatically, the chances of website crashes are greatly reduced.

Improving User Experience

- Faster page loading improves customer satisfaction.
- Customers can browse products, add items to the cart, and complete payments without delays.
- Smooth performance encourages more purchases and reduces cart abandonment.

Role of Monitoring Tools

Monitoring tools continuously observe the health and performance of the cloud infrastructure.

They monitor:

- CPU and memory usage
- Number of active users
- Network traffic
- Application response time

When predefined thresholds are exceeded:

- Alerts are sent to administrators.
- Auto-scaling is triggered to launch additional servers.
- Performance dashboards help identify issues before customers are affected.

This proactive monitoring ensures that the system remains reliable during periods of high demand.

Real-World E-Commerce Scenario

Consider an online shopping platform during the Amazon Great Indian Festival **or** Flipkart Big Billion Days sale.

- At the start of the sale, millions of customers visit the website simultaneously.
- The load balancer distributes customer requests across multiple application servers.
- As traffic continues to grow, auto-scaling automatically launches additional servers to handle the increased workload.
- Monitoring tools detect high CPU usage, memory utilization, and response times, ensuring that resources are added before performance degrades.
- If one server fails, customer traffic is immediately redirected to healthy servers without interrupting the shopping experience.

As a result, customers can browse products, place orders, and complete payments smoothly without experiencing slow pages or website crashes.

2. **An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.**

How Cloud Storage Ensures Durability and Data Redundancy

Organizations generate large amounts of backup data, including databases, documents, application files, and system images. Cloud storage provides a secure and reliable way to store this data while protecting it from hardware failures and disasters.

Cloud storage ensures durability and data redundancy through the following methods:

- Data is automatically stored in multiple copies across different storage devices.
- Cloud providers replicate backup data across multiple servers and, in many cases, across different availability zones or regions.
- If one storage device fails, another copy is immediately available.
- Regular data integrity checks detect and repair corrupted files automatically.
- Backup versions are maintained so deleted or modified files can be restored when needed.

As a result, organizations can rely on cloud storage to keep backup data safe and available even during hardware failures.

Protecting Backup Data with Encryption and Access Control

Sensitive backup data must be protected from unauthorized access. Cloud providers achieve this using encryption and access control.

Encryption

Encryption converts readable data into an unreadable format that can only be accessed with the correct encryption key.

- **Data at Rest:** Backup files stored in the cloud are encrypted using strong algorithms such as AES-256.
- **Data in Transit:** Data transferred between the organization and the cloud is protected using secure protocols such as TLS/SSL.
- Encryption keys are securely managed using Key Management Services (KMS) or Hardware Security Modules (HSMs).

Access Control

Access control ensures that only authorized users can access backup data.

- Use Identity and Access Management (IAM) to define user permissions.
- Implement Role-Based Access Control (RBAC) so users access only the backups relevant to their responsibilities.
- Enable Multi-Factor Authentication (MFA) for additional security.
- Maintain audit logs to track who accessed or modified backup data.

These measures protect confidential information from unauthorized users and cyber threats.

Cost Advantages Compared to Traditional Storage Systems

Cloud storage is generally more cost-effective than maintaining traditional on-premises backup systems.

Advantages include:

- **Pay-as-you-go pricing:** Organizations pay only for the storage they actually use.
- **No hardware investment:** There is no need to purchase expensive storage servers or backup devices.
- **Reduced maintenance costs:** Cloud providers manage hardware, upgrades, and maintenance.
- **Automatic scalability:** Storage capacity can be increased or decreased as needed without purchasing new equipment.
- **Lower operational costs:** Reduced power consumption, cooling, and physical space requirements.
- **Lifecycle management:** Older backups can be moved automatically to lower-cost archival storage, reducing overall expenses.

Overall, cloud storage provides flexibility while minimizing capital and maintenance costs.

Disaster Recovery Use Case

Scenario:

A company stores its business applications and customer databases in the cloud. One day, a ransomware attack encrypts the primary database, making it inaccessible.

Disaster Recovery Process:

1. The organization identifies the attack and isolates the affected systems.
2. Cloud backup services retrieve the most recent clean backup.
3. The database is restored from encrypted cloud storage.
4. Replicated backup data from another region is used if the primary region is unavailable.
5. Business applications are restarted, allowing employees and customers to continue using the services with minimal downtime.

This process enables the organization to recover important data quickly and resume operations without significant data loss.

- 3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.**

How Content Delivery Networks (CDNs) Improve Performance and Reduce Latency

A mobile app company with users across the world needs to deliver content quickly and reliably. If all users access data from a single server, people located far from the server may experience slow loading times and delays. A Content Delivery Network (CDN) solves this problem by storing copies of content on multiple servers located around the world.

A CDN improves performance by:

- Storing frequently accessed content closer to users.
- Reducing the distance data travels over the internet.
- Decreasing network latency and improving response time.
- Reducing the workload on the main application server.
- Delivering content faster during periods of high traffic.

As a result, users experience faster app loading, smoother media playback, and quicker downloads.

How Edge Servers Enhance User Experience Across Regions

Edge servers are servers located at different geographical locations that are part of a CDN. They deliver content from the nearest location instead of the central server.

Edge servers improve the user experience by:

- Providing faster access to images, videos, and app resources.
- Reducing buffering during video streaming.
- Lowering latency for online gaming and real-time applications.
- Handling large numbers of user requests without overloading the origin server.
- Improving application availability even if the main server experiences heavy traffic.

Example:

- A user in India receives content from an edge server in India.
- A user in Europe receives the same content from a nearby European edge server.

This significantly reduces response time for users worldwide.

How Cloud Providers Support Global Availability

Cloud providers maintain data centers and CDN edge locations across many countries, enabling applications to serve users globally.

They support global availability by:

- Deploying applications in multiple cloud regions.
- Providing CDN services to cache and deliver content from edge locations.

- Using global load balancers to direct users to the nearest healthy server.
- Replicating application data across multiple regions for high availability.
- Offering automatic failover so applications remain available even if one region experiences an outage.
- Using auto-scaling to handle sudden increases in user traffic.

These services ensure that applications remain fast, reliable, and accessible from anywhere in the world.

Real-Time Streaming or Gaming Example

Example: Online Video Streaming Platform

Consider a video streaming platform such as Netflix or YouTube.

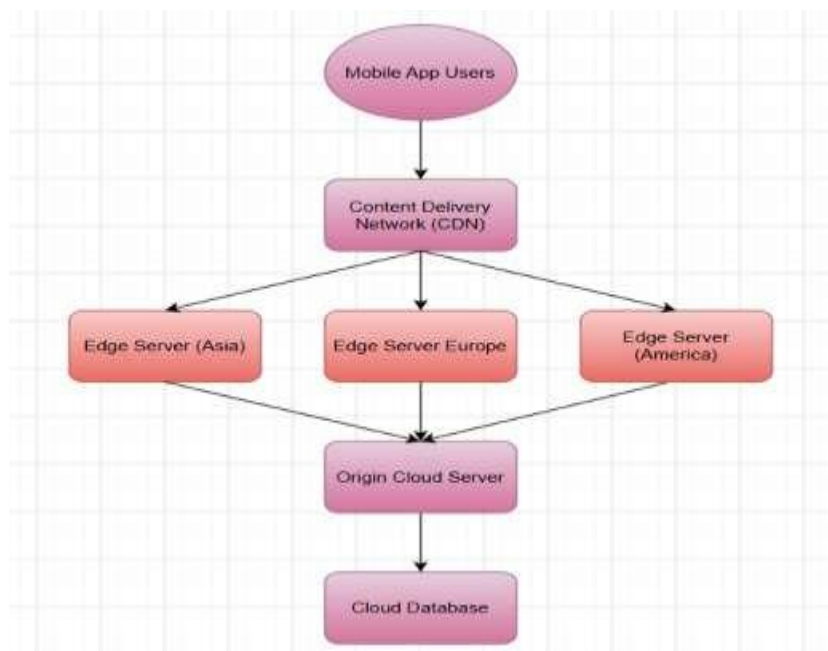
- A user in India opens the app to watch a movie.
- Instead of downloading the video from a server in another country, the CDN delivers the video from the nearest edge server.
- This reduces buffering and allows the video to start playing almost immediately.
- During popular live events, millions of users watch simultaneously. The CDN distributes traffic across many edge servers, preventing the main server from becoming overloaded.

Example: Online Multiplayer Gaming

In an online multiplayer game:

- Edge servers reduce communication delays between players and game servers.
- Lower latency results in smoother gameplay, faster response to player actions, and a better gaming experience.
- Cloud providers automatically scale gaming servers during tournaments or new game launches to support thousands of players at the same time.

Architecture Diagram:



4. **A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.**

How Cloud Providers Support Regulatory Compliance and Auditing

Financial institutions handle highly sensitive customer information, including account details, credit card data, and transaction records. Therefore, they must comply with strict regulations and security standards. Cloud providers offer several services to help organizations meet these compliance requirements.

Cloud providers support compliance by:

- Providing infrastructure that complies with standards such as ISO 27001, PCI DSS, SOC 2, and GDPR.
- Offering built-in audit logs that record every user activity and system event.
- Maintaining detailed documentation and compliance reports for regulatory audits.
- Providing secure backup and disaster recovery services.
- Supporting data retention policies required by financial regulations.
- Performing regular security assessments and vulnerability management.

These features help organizations demonstrate compliance during internal and external audits.

Role of Logging, Monitoring, and Encryption in Security

Logging

Logging records all activities performed within the cloud environment.

Examples include:

- User login attempts
- Financial transactions
- Configuration changes
- Data access events
- Failed authentication attempts

Logs help security teams investigate incidents, detect suspicious activities, and maintain audit trails.

Monitoring

Monitoring continuously checks the health and security of cloud resources.

Monitoring tools can:

- Detect unusual login attempts.
- Identify abnormal transaction patterns.
- Monitor server performance and application health.
- Generate alerts when suspicious activities occur.
- Help administrators respond quickly to potential cyberattacks.

Continuous monitoring improves system reliability and reduces security risks.

Encryption

Encryption protects sensitive financial information from unauthorized access.

Cloud providers implement:

- **Encryption at Rest:** Customer data stored in databases and cloud storage is encrypted using strong algorithms such as AES-256.
- **Encryption in Transit:** Data moving between customers, applications, and cloud servers is protected using TLS/SSL.
- Secure Key Management Services (KMS) to safely manage encryption keys.
- Regular encryption key rotation for enhanced security.

Even if data is intercepted, encryption prevents unauthorized users from reading it.

Ensuring Data Sovereignty Requirements

Data sovereignty means that an organization's data must remain within a specific country or geographic region according to local laws and regulations.

A financial institution can ensure data sovereignty by:

- Selecting cloud regions located within the required country.
- Storing customer databases only in approved geographic locations.
- Configuring backup and disaster recovery systems to use compliant regions.
- Restricting cross-border data transfers unless legally permitted.
- Applying location-based access policies for sensitive information.
- Regularly auditing storage locations to verify compliance.

This helps the organization meet legal and regulatory requirements while protecting customer data.

Example: Sensitive Financial Transactions

Scenario:

A customer transfers ₹5,00,000 using an online banking application.

The cloud environment handles the transaction as follows:

1. The customer logs in using Multi-Factor Authentication (MFA).
2. The banking application verifies the user's identity and account balance.
3. Every step of the transaction is recorded in secure audit logs.
4. Monitoring tools continuously check for suspicious activities, such as unusually large transfers or repeated failed login attempts.
5. Customer account information and transaction records are stored in encrypted databases located within the country's approved cloud region.
6. Security teams receive immediate alerts if any unusual behavior is detected.

This process ensures that the transaction is secure, traceable, and compliant with financial regulations.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Difference Between Containers and Virtual Machines

A development team wants to deploy applications quickly and efficiently. Containerization is a modern approach that packages an application along with its dependencies, allowing it to run consistently across different environments. Unlike Virtual Machines (VMs), containers share the host operating system, making them lightweight and faster.

Feature	Containers	Virtual Machines
Operating System	Share the host OS	Each VM has its own guest OS
Startup Time	Starts within seconds	Takes several minutes to boot
Resource Usage	Lightweight and uses less CPU and memory	Requires more CPU, memory, and storage
Performance	Faster due to lower overhead	Slightly slower because of virtualization
Portability	Easily runs across different environments	Less portable due to OS dependencies
Scalability	Can be scaled quickly	Scaling is slower because entire VMs must be created

Because containers consume fewer resources and start almost instantly, they are ideal for cloud-native applications.

How Kubernetes Simplifies Deployment and Scaling

When an application consists of many containers, managing them manually becomes difficult. Kubernetes is a container orchestration platform that automates deployment, scaling, and management.

Kubernetes simplifies deployment by:

- Automatically deploying containers across available servers.
- Restarting failed containers without manual intervention.
- Performing rolling updates without interrupting users.
- Managing container networking and service discovery.
- Distributing workloads evenly across cluster nodes.

Automatic Scaling

Kubernetes also supports automatic scaling.

- Horizontal Pod Autoscaler (HPA) increases or decreases the number of containers based on CPU or memory usage.
- Cluster Autoscaler adds or removes worker nodes when more computing power is required.
- During high traffic, new containers are created automatically.
- When traffic decreases, unused containers are removed to reduce costs.

This ensures high availability while optimizing resource utilization.

Portability Across Different Cloud Environments

One of the biggest advantages of containers is portability.

Containers package the application together with its libraries, runtime, and dependencies. As a result, the same container image can run without modification on:

- Public cloud platforms
- Private cloud environments
- Hybrid cloud infrastructures
- Local development machines

This allows organizations to:

- Avoid vendor lock-in.
- Move applications easily between cloud providers.
- Maintain consistent application behavior in development, testing, and production.
- Simplify disaster recovery and multi-cloud deployments.

Portability enables organizations to deploy applications wherever required without changing the application code.

Example of a Microservices-Based Application

Example: Online Food Delivery Application

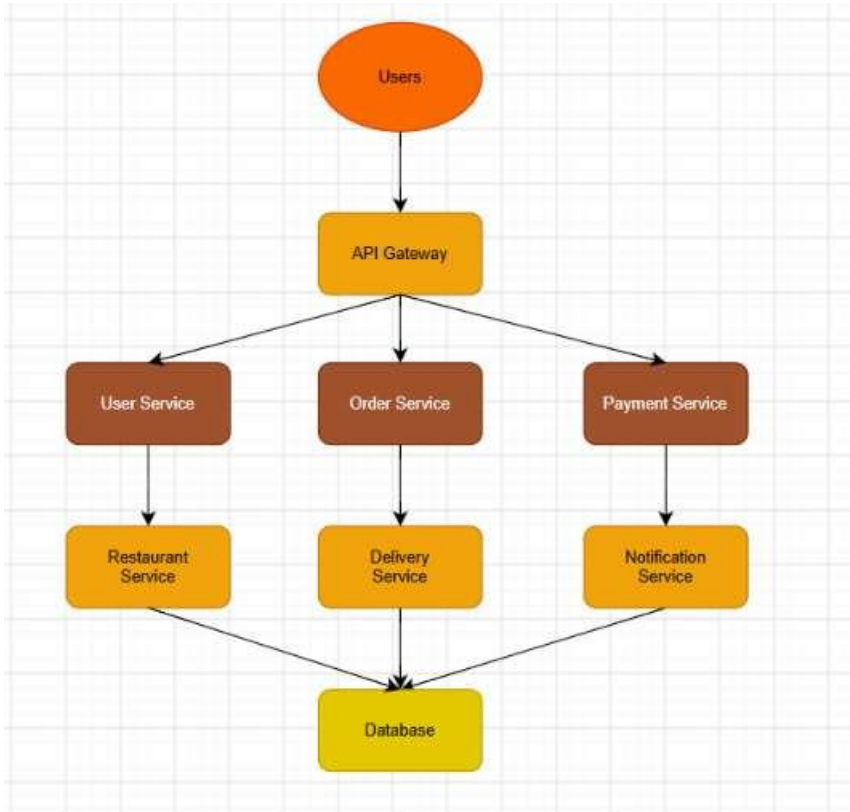
Instead of building one large application, the system is divided into several independent microservices.

Microservices include:

- **User Service** – Manages user registration and login.
- **Restaurant Service** – Displays restaurant information and menus.
- **Order Service** – Processes customer orders.
- **Payment Service** – Handles online payments securely.
- **Delivery Service** – Tracks delivery partners and order status.
- **Notification Service** – Sends SMS, email, or push notifications.

Each microservice runs inside its own container and can be updated or scaled independently.

Architecture Diagram:



Example Scenario:

During lunch hours, thousands of customers place food orders simultaneously.

- Kubernetes automatically creates more Order Service and Payment Service containers.
- The User Service continues running normally because it has less traffic.
- If a payment container fails, Kubernetes immediately starts a new one.
- Once demand decreases, extra containers are removed automatically.

This ensures smooth performance without wasting computing resources.